MPL: Mine the Pseudo Labels for Semi-Supervised Temporal Action Localization

Peng Lu[a,b,1], Jinjin Jin[a,b,2], Jiemeng Tao[a,b,3], Shaofeng Liu[a,b,1], Yangyang Wu[c,1], Jianfeng Zhang[a,1], Peijian Cao[a,b,1,*]

[a] *Beijing Life Science Academy, , Beijing, 102200, China*
[b] *Zhengzhou Tobacco Research Institute of CNTC, Zhengzhou, 450001, China*
[c] *Center for Data Science, Zhejiang University, 866 Yuhangtang Road, Hangzhou, 310058, China*

Abstract

Semi-Supervised Temporal Action Localization (SS-TAL) has attracted increasing attention for its ability to utilize unlabeled data. Most of the existing methods adopt pseudo labels to improve the utilization of unlabeled data. However, these methods neglect the unbalanced distribution of label noise in various samples and treat all pseudo labels equally. In this paper, we propose a novel framework called Mine the Pseudo Labels (MPL) to address this issue. Specifically, we first propose a method called pseudo-label confidence ranking which consists of three metrics to reliably measure the sample-wise quality of pseudo labels. Further, we design the pseudo-label adaptive correction module to refine the low confidence pseudo labels by estimating and rectifying the inherent bias of the temporal localization of the model. In addition, we introduce the dual-level consistency learning with a Gaussian sampling scaling strategy, to train the model to learn scale-invariant and

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*Corresponding authors

Email addresses: penglu2004@hotmail.com (Peng Lu), jinjingjing1218@126.com (Jinjin Jin), taojiemeng_zz@163.com (Jiemeng Tao), sfliu@126.com (Shaofeng Liu), zjuwuyy@zju.edu.cn (Yangyang Wu), zhangjianfeng83@outlook.com (Jianfeng Zhang), peijiancao@163.com (Peijian Cao)

[1] First author
[2] Project administrator
[3] Algorithm design consultant
[4] Experimental validation

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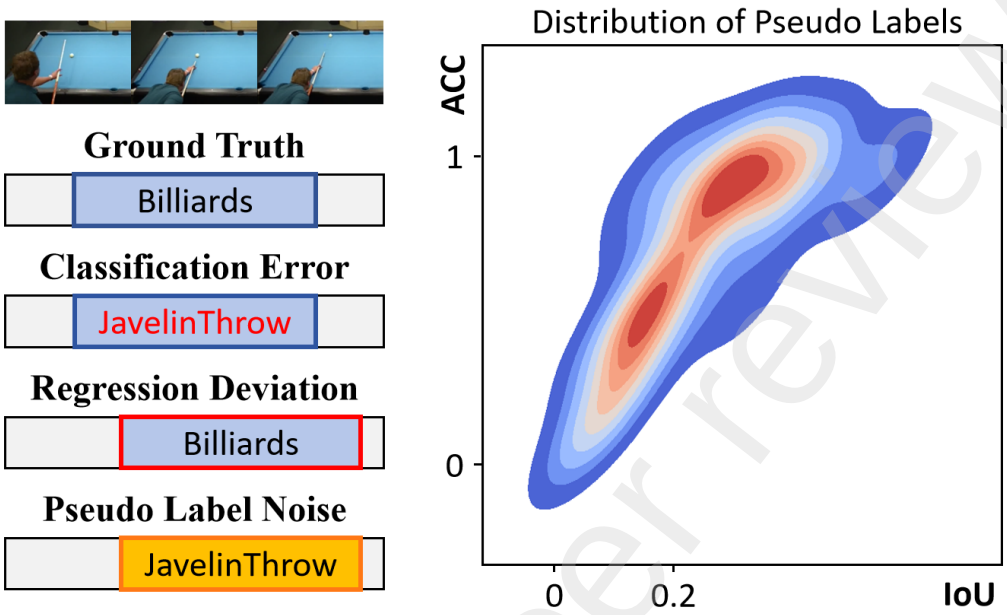

Figure 1: The illustration of unbalanced distribution of pseudo label noise among samples in THUMOS14 training set, with the distribution given by 2-D kernel density estimation. Noise denotes the difference of pseudo labels from the real labels. A higher ACC indicates a more accurate judgment of action types, while a larger IoU signifies a greater precision in the time positioning of the action.

noise-unaware feature representation. Extensive experiments conducted on the benchmark datasets THUMOS14 and ActivityNet v1.3 demonstrate that our method outperforms previous state-of-the-art methods at 10% label rate and 60% label rate.

Keywords: temporal action localization; semi-supervised learning; pseudo label

1. Introduction

Temporal Action Localization (TAL) aims to identify and locate specific actions from consecutive video frames or snippets [1]. In recent years, with the rapid development of deep learning and the fast growth of computing power, TAL has made significant progress. However, in practical applications, annotating video data is an extremely time-consuming and costly task, which greatly limits the promotion and application of TAL technology.

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Therefore, studying how to effectively utilize unlabeled video data for learning with a limited amount of labeled video data, namely semi-supervised temporal action localization (SS-TAL), holds considerable theoretical significance and practical value.

Existing semi-supervised learning methods [2, 3, 4, 5, 6] rely on consistency learning and self-training. Majority of them [3, 4, 6] employ the teacher-student framework in which pseudo labels are generated from unlabeled data and subsequently used as supervisory information for training. Besides, [5] proposes a single-stage architecture to eliminate the proposal error propagation. [6] combines the class score with boundary uncertainty and balances the distribution of samples across different categories. Nevertheless, these methods overlook the significant quality differences among pseudo-labels of different samples, as shown in Figure 1. Equally treating pseudo-labels that mixed in quality during training may inadvertently reinforce cognitive errors within the model, leading to sub-optimal model performance. In addition, we also find that there is an inherent bias in the model's localization, with temporal intervals typically being either earlier or later on average.

In response to the above issues, we propose a novel SS-TAL framework called **M**ine the **P**seudo **L**abels (MPL) which comprises three key components: pseudo-label confidence ranking, pseudo-label adaptive correction and dual-level consistency learning.

First, for pseudo-label confidence ranking, we merge three useful metrics into one overall confidence score for each pseudo-labeled sample, inspired by [7]. This approach for pseudo-label evaluation aids in distinguishing pseudo-labels of varying qualities throughout the training process, effectively mitigating the adverse effects of low-quality pseudo-labels on model training.

Then, for pseudo-label adaptive correction, we randomly sample a set of neighboring intervals around each predicted time interval and perform re-inference on the features corresponding to these neighboring intervals. The results of re-inference show significant temporal shifts, which we attribute to the inherent bias of the model's temporal localization. By correcting this bias in the original predicted time interval, we can achieve more accurate temporal positioning.

Finally, for dual-level consistency learning, we introduce scale-invariant feature learning specifically tailored for videos. Taking into account the scale variability of action instances, we devise a scaling rate sampling strategy to augment the feature, enabling the model to extract scale-insensitive feature

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representations. This enhances the model's performance in action detection under complex video environments.

We evaluate our method on two benchmark datasets THUMOS14 and ActivityNet v1.3, aligning with previous researches. Our method achieves outstanding performance. Specifically, on THUMOS14, our method demonstrates an improvement of 1.7% and 1.1% in average mAP compared to the previous state-of-the-art methods at 10% and 60% label rates, respectively. On the ActivityNet v1.3 dataset, our method shows an improvement of 0.5% and 0.5% in average mAP at the corresponding label rates.

In summary, our contribution is threefold:

- We propose Pseudo-label Confidence Ranking and Pseudo-label Adaptive Correction to effectively estimate the relative quality of pseudo labels and significantly enhance the localization precision of low-confidence pseudo labels.
- We propose Dual-level Consistency Learning which incorporates a Gaussian sampling scaling strategy to enable the model to learn noise-unaware and scale-insensitive feature representation.
- Extensive experiments conducted on two major benchmarks, THUMOS14 and ActivityNet v1.3, demonstrate that MPL effectively utilizes unlabeled data to enhance temporal action localization performance, outperforming prior state-of-the-art methods.

2. Related Work

2.1. Temporal Action Localization

Temporal Action Localization (TAL) [1] methods are usually divided into two-stage, single-stage and anchor-free methods. Two-stage methods [8, 9, 10, 11, 12, 13] first generate candidate instance-wise action proposals, then perform boundary regression and action recognition on each proposal. The proposals may be generated by fixed anchors with single or multiple sizes [14, 15, 16, 17], or through direct boundary regression [18, 19]. Two-stage frameworks often feature elaborate fusion but suffer from complicated design and proposal error propagation. Single-stage methods [8, 20, 21, 22] directly predict temporal boundaries and action categories for each instance. Anchor-free methods [23, 24, 25, 26, 27] perform action boundary regression without using default anchors. Because of their simple yet efficient frameworks, recent

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anchor-free models, such as ActionFormer [26], TriDet [27] and MSTN [28], achieve surprising performance in the TAL task.

2.2. Semi-Supervised Learning

Semi-Supervised Learning (SSL) [29] methods can overall be divided into generative [30, 31], graph-based [32, 33], consistency regularization [34, 35, 36, 37, 38, 39], and pseudo-labeling methods [40, 41, 42, 43]. SSL finds numerous applications in visual tasks, including image classification, object detection, semantic segmentation. Many SSL methods in these tasks employ pseudo-label techniques [44, 45, 46, 47] and consistency learning framework [38, 39]. Mean Teacher [37] averages model weights by EMA to achieve better performance without changing network architecture. Active Teacher [7] evaluates unlabeled samples based on three key criteria and maximizes the utilization of limited label information. PseCo [48] introduces multi-view scale-invariant learning in object detection. Zhang *et al.* [49] and PCL [50] try to refine pseudo labels for more reliable training.

2.3. Semi-Supervised TAL

Existing Semi-Supervised Temporal Action Localization (SS-TAL) methods can be divided into two branches. The first branch focuses on consistency learning and self-training [2, 3, 4]. KFC [2] requires the single model to make consistent predictions for features with added spatial and temporal perturbation. Subsequently, Ji *et al.* [3] combine the teacher-student framework with SS-TAL, proposing time warping and time masking strategy. Further, SSTAP [4] introduces self-supervised module which learns relation-aware feature by feature reconsturction and clip-order prediction training. The other branch of SS-TAL studies post-processing methods [5, 6]. SPOT [5] proposes a single-stage TAL solution to alleviate error propagation between action proposal generation and classification. NPL [6] combines class score with boundary ambiguity and limits the maximum number of predictions in each video, providing another perspective on SS-TAL. These researches achieve good results in experiments. However, the methods using pseudo labels do not pay enough attention to the pseudo labels themselves which contain valuable video knowledge. Therefore, we propose our SS-TAL framework MPL which focuses on dealing with the noise within pseudo labels. Our research on utilizing and improving pseudo labels can fill this gap in the field of SS-TAL.

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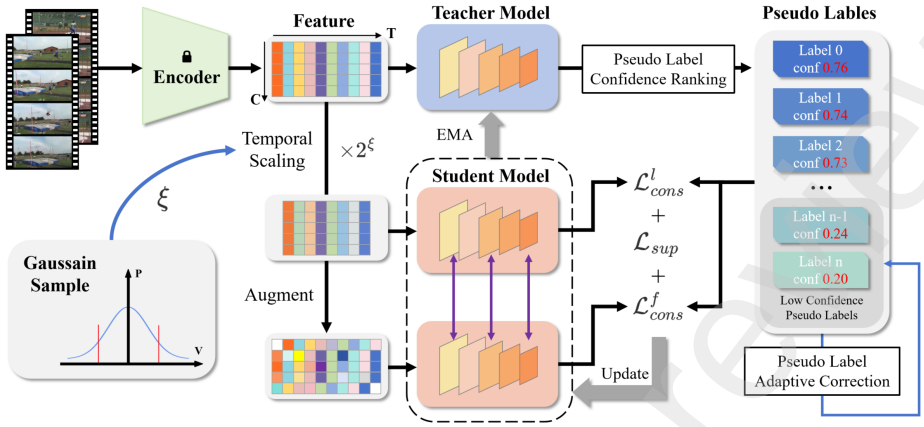

Figure 2: An overview of the proposed MPL framework in semi-supervised training phase. First, the feature extractor encodes raw videos to generate video features. Subsequently, the teacher model uses features to obtain pseudo labels and confidence scores. Pseudo labels with low confidence undergo adaptive correction, as shown in Figure 3. For label-level consistency, temporal scaling is employed on features, which is intended to enable the model to extract scale-invariant features for actions of different sizes. For feature-level consistency, noise disturbances are added to the unscaled features, and under the supervision of pseudo labels, the student model is trained to generate consistent predictions for the augmented features.

3. Method

3.1. Preliminary

Problem Definition. The task of SS-TAL is to improve the performance of TAL network by fully utilizing both labeled videos $\{X_i^l\}_{i=1}^{N_l}$ and unlabeled videos $\{X_i^u\}_{i=1}^{N_u}$. Each action segment in the labeled video is annotated with (s, e, c), where s, e and c refer to start time, end time, and category of this action instance. The performance of TAL network is reflected in its accuracy in temporal localization $(\hat{s}, \hat{e})$ and classification $\hat{c}$.

Overview. Our method is based on teacher-student framework and self-training paradigm. We divide the whole training process into two phases. In the pre-training phase, we obtain a pre-trained model trained with labeled data and supervised loss, to initialize the teacher and student model. In the semi-supervised training phase, which is shown in Figure 2, teacher model generates dense predictions along the time dimension for input video sample. Then, a low threshold is applied to the class scores of these predictions, and Non-Maximum Suppression (NMS) is used to obtain pseudo

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label $\{s_j, e_j, c_j\}_{j=1}^{J}$, where J denoting the number of valid action predictions in a specific video (in this paper, the letter j uniformly refers to the action prediction id in a video sample).

The proposed method is divided into three parts, namely **P**seudo-label **C**onfidence **R**anking (PCR), **P**seudo-label **A**daptive **C**orrection (PAC), and **D**ual-level **C**onsistency **L**earning (DCL). PCR evaluates the confidence of all pseudo labels on unlabeled samples and sorts them by calculated comprehensive confidence. PAC is tailored for SS-TAL to improve the precision of temporal intervals in pseudo labels with low confidence. DCL helps the model learn scale-agnostic and robust-to-noise feature representation, and in this section we design a simple but effective temporal scaling strategy.

3.2. Pseudo-label Confidence Ranking

ActiveTeacher [7] proposes an algorithm that estimates value of unlabeled samples for semi-supervised objection detection (SSOD), which evaluates pseudo labels from three perspectives: difficulty, information content, and category diversity. Based on the evaluation results, the pseudo-labeled videos are adaptively sampled, achieving good results in SSOD. However, this method of evaluating value for OD is not well suited for TAL task as we find that high-value samples implied by pseudo labels are difficult to learn and hence cause sub-optimal performance. We reasonably speculate that this phenomenon is caused by different distributions of datasets. In the COCO 2017 validation set, the average number of objects and categories contained in a single image is 2.95 and 7.43 respectively. By contrast, in THUMOS14 training set, a training sample contains about 15.0 action instances and 1.1 action categories. And in ActivityNet1.3, average training sample contains only 1.5 action instances and 1 action category.

Aforementioned information indicates that if a video sample seems difficult, informative and diverse in categories, it is more likely accompanied by uncertain identification or incorrect inclusion of categories. This finding provides us with a new perspective. Therefore, we propose Pseudo-label Confidence Ranking (PCR), which evaluates confidence of pseudo labels from both action and video levels, taking into account the boundary ambiguity.

Action-wise class uncertainty. This uncertainty value is quantified using information entropy, which is calculated based on the category output values provided by the detection head. The higher the information entropy, the stronger the uncertainty of the model in predicting categories. Specifi-

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cally, it is represented as:

$$U_{cls}^{A} = \frac{1}{J}\sum_{j=1}^{J}\sum_{i=1}^{N_{cls}}(-\hat{p}_{ji}\log\hat{p}_{ji}) \tag{1}$$

where $\hat{p}_{ji}$ represents the output value of the classifier of the detector for the i-th category of the j-th action instance prediction. N_{cls} denotes the number of activity categories. The predictions are obtained from dense outputs after applying both NMS and a score threshold filter.

Sample-wise class uncertainty. The first metric measures action-wise confidence. Hence we further design a indicator to measure video-level uncertainty of classification. Specifically, count whether various actions appear in a video to form a vector of length N_{cls} and then get its module length:

$$U_{cls}^{V} = \left|\left|\{\mathbb{1}_{[N_i>0]}\{1-\mathrm{MAX}_{j=1}^{J} sigmoid(\hat{p}_{ji})\}\}_{i=1}^{N_{cls}}\right|\right| \tag{2}$$

in which N_i denotes the frequency of i-th category in all predictions and N_{cls} refers to the number of categories. $\mathbb{1}_{[\cdot]}$ is indicator function which returns 1 if the condition is met, and 0 otherwise.

Regression uncertainty. For dense predictions of a video, after filtering through NMS and basic score thresholds, J action predictions will be obtained. For each interval $(\hat{s}_j, \hat{e}_j)$ with local maximum score, using an IoU threshold of 0.4 by default, we find a set of adjacent intervals $[\tilde{\mathbf{s}}_j, \tilde{\mathbf{e}}_j] \in \mathbb{R}^{K\times 2}$ predicted with the same action type, where K denotes the number of adjacent intervals to the j-th action. Then, uncertainty is measured by the average standard deviation of the boundary:

$$U_{reg} = \frac{1}{J}\sum_{j=1}^{J}\frac{\sigma(\tilde{\mathbf{s}}_j)+\sigma(\tilde{\mathbf{e}}_j)}{\hat{e}_j-\hat{s}_j} \tag{3}$$

where $\sigma(\cdot)$ denotes standard deviation calculation. To normalize the uncertainty measure, the standard deviation is divided by $\hat{e}_j - \hat{s}_j$ which represents the length of the j-th interval.

Unification of metrics. The above introduced metrics for sorting video samples to some extent indicate the confidence level of pseudo labels. However, there is a significant difference in the numerical magnitude of these indicators, with U_{cls}^{A} typically ranging from 1 to 3, U_{cls}^{V} almost always smaller than 1 and U_{reg} usually varying between 0.4 and 0.8. Moreover, the differences in video sample distribution and model variations may exacerbate this

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issue. To appropriately integrate these three metrics, we apply max normalization and employ L-p norm to avoid excessive influence from any single indicator:

$$conf = 1 - \left\| \left\{ \frac{U^A_{cls}}{U^A_{max_cls}}, \frac{U^V_{cls}}{U^V_{max_cls}}, \frac{U_{reg}}{U_{max_reg}} \right\} \right\|_p \Big/ 3^{\frac{1}{p}} \tag{4}$$

$U^A_{max_cls}$, $U^V_{max_cls}$ and U_{max_reg} represent the maximum value of each metric in all samples. p is set to 1 by default.

After calculating the confidence of all pseudo labels and ranking them in descending order, the top 50% of pseudo labels are considered to be high-confidence labels, while the rest are considered to have low reliability. Our ablation study demonstrates the effectiveness of each metric and comprehensive confidence, and the experiment results are shown in Table 4.

3.3. Pseudo-label Adaptive Correction

As shown in Table 3, samples with high confidence are easier for the model to learn from compared to those with low confidence. However, this does not necessarily mean that unreliable pseudo labels should be discarded. In fact, although there is a lot of noise in pseudo labels with low confidence, these ambiguously predicted temporal intervals are likely to be near the ground truth. In addition, we find that the predicted intervals are often not centered on the real labels. To be specific, the expected position of the prediction intervals is biased, either early or late. For pseudo interval $(\hat{s}, \hat{e})$ and real interval (s^{gt}, e^{gt}):

$$E(\frac{1}{n}\sum_{i=1}^{n}\frac{s_i^{gt} - \hat{s}_i}{\hat{e}_i - \hat{s}_i}) \neq 0, \quad E(\frac{1}{n}\sum_{i=1}^{n}\frac{e_i^{gt} - \hat{e}_i}{\hat{e}_i - \hat{s}_i}) \neq 0 \tag{5}$$

Based on this finding, we design Pseudo-label Adaptive Correction (PAC), which is illustrated in Figure 3. Specifically, we first randomly sample some areas in the vicinity of the prediction interval $(\hat{s}, \hat{e})$ to form a set of adjacent intervals:

$$\begin{gathered} \epsilon \sim N(0, \sigma_{psc}), \epsilon \in \mathbb{R}^{N_{psc} \times 2} \\ \{\tilde{s}_i, \tilde{e}_i\}_{i=1}^{N_{psc}} = \epsilon \times (\hat{e} - \hat{s}) + \text{Repeat}([\hat{s}, \hat{e}], N_{psc}) \end{gathered} \tag{6}$$

where the repeat function expands the pseudo interval in 0-th dimension N_{psc} times and σ_{psc} controls the length of potential sampling region. Then, we use the teacher model to infer the corresponding region features from

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this set of intervals in order to obtain a set of jittered predictions, which are weighted by class-confidence:

$$(\overline{s}, \overline{e}) = \frac{1}{\sum_{i=1}^{N_{psc}} \hat{q}_i} \sum_{i=1}^{N_{psc}} \hat{q}_i \cdot f_{reg}(\mathcal{F}_{:,\tilde{s}_i:\tilde{e}_i} \mid \theta_t) \tag{7}$$

where $\hat{q}_i$ refers to classification score of the i-th prediction, $f_{reg}(\cdot \mid \theta_t)$ represents the teacher model's inference process for determining the time interval. $(\overline{s_i}, \overline{e_i})$ denotes a weighted and secondary-inferred prediction.

Subsequently, we estimate the inherent localization bias of model by calculating:

$$\overline{\delta}_s = \frac{1}{n} \sum_{i=1}^{n} \frac{\overline{s}_i - \hat{s}_i}{\hat{e}_i - \hat{s}_i}, \quad \overline{\delta}_e = \frac{1}{n} \sum_{i=1}^{n} \frac{\overline{e}_i - \hat{e}_i}{\hat{e}_i - \hat{s}_i} \tag{8}$$

Then, use this estimated bias to correct initial pseudo label:

$$(s^{psc}, e^{psc}) = (\hat{s}, \hat{e}) - \alpha_{psc}(\hat{e} - \hat{s})(\overline{\delta}_s, \overline{\delta}_e) \tag{9}$$

This process does not require additional parameters by reusing the components of teacher model. In actual implementation, the inherent bias is estimated from all low-confidence samples rather than a single sample. The experimental results in Table 4 and Table 6 demonstrate the effectiveness of pseudo label adaptive correcting.

3.4. Dual-level Consistency Learning

Previous researches [2, 4, 5] propose many consistency learning methods. For instance, time shift/flip and Gaussian noise are used to augment features to relieve the impact of noise, with time masking and order disturbance performed for self-supervised learning in spatial-temporal relationships. In object detection, it is common to downsample images to different scales and require consistency at the feature level. However, simply performing random or fixed-magnitude downsampling for scale-invariant consistency training in video tasks does not achieve good results, and in fact performs worse.

We reasonably speculate that this is due to the fact that vanilla downsampling strategy alters the distribution of sample data, causing the network to favor shorter interval predictions. This artificially introduces disruptive noise into the process of simple downsampling. To address this drawback and enable the network to better learn scale-invariant features from action instances, we propose a simple but effective Gaussian sampling scaling strategy

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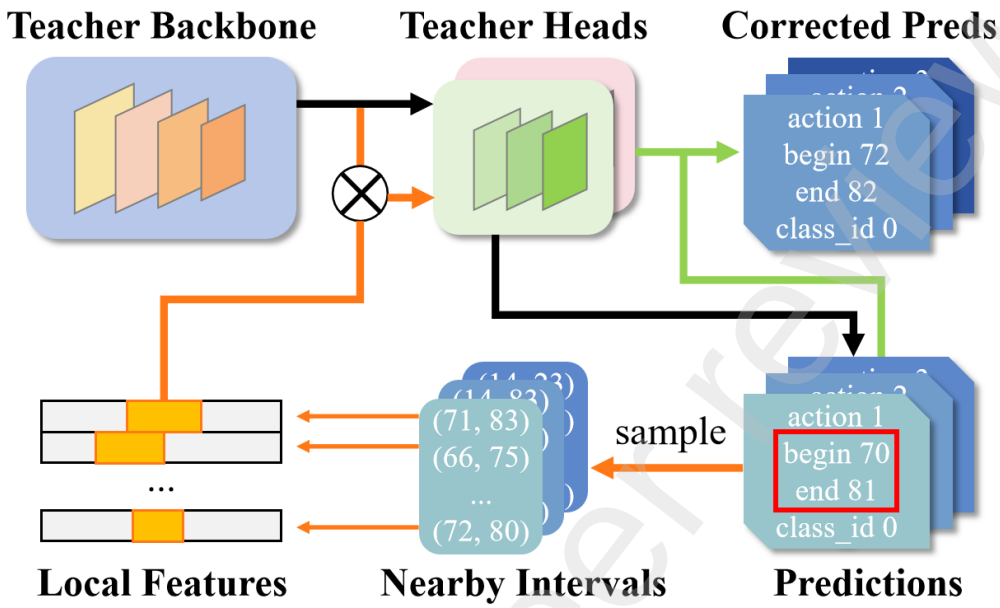

Figure 3: The illustration of Pseudo-label Adaptive Correction. In the first period, as indicated by black lines, the outputs from teacher model's backbone is sent to heads to get preliminary predictions. In the second period, as shown by orange lines, nearby intervals are generated by adding Gaussian noise to each predicted interval, then the inherent localization bias between the predictions and the re-predictions is estimated. Finally, as indicated by green line, initial predictions are corrected with the estimated bias.

for SS-TAL, along with a dual-level consistency approach for scale-invariant and noise-unaware learning. This part aims to mitigate the negative impact of vanilla downsampling and enhance the network's ability to capture meaningful temporal patterns across different scales.

Specifically, we first perform random Gaussian sampling to obtain scaling factors around 0. Subsequently, we map these factors to an exponential space to obtain scaling ratios. This process ensures that the scaling ratios are symmetrically distributed in the logarithmic space and within a reasonable range. By doing so, controlled yet diverse scale perturbations are introduced to the input data:

$$\gamma = \mathbb{1}_{[|\xi|\leq 1]} 2^{\xi}, \quad \xi \sim N(0, \sigma) \tag{10}$$

where γ denotes the scaling ratio, and σ is a parameter set to 1 by default. We re-sample the ξ if ξ is not in the interval [-1, 1] to ensure the scaling ratio

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γ within a reasonable range [0.5, 2].

After obtaining γ, we scale the feature $\mathcal{F} \in \mathbb{R}^{C \times T}$ in the time dimension by linear interpolation:

$$\mathcal{F}_\gamma = \mathcal{F} \times \mathbf{Q} \tag{11}$$

where $\mathbf{Q} \in \mathbb{R}^{T \times T'}$ refers to the transformation matrix for linear interpolation, $T' = \lfloor \gamma \times T \rfloor$ and $\mathcal{F}_\gamma \in \mathbb{R}^{C \times T'}$ denotes scaled feature.

Following that, send the initial feature $\mathcal{F}$ and scaled feature $\mathcal{F}_\gamma$ to backbone network to obtain their further products, $\mathcal{F}' \in \mathbb{R}^{C_b \times T}$ and $\mathcal{F}'_\gamma \in \mathbb{R}^{C_b \times T'}$, correspondingly. Finally, we perform the same scaling operation on $\mathcal{F}'$ as above, to make it align with $\mathcal{F}'_\gamma$ in time dimension to calculate the scale-invariant consistency loss:

$$\mathcal{L}^l_{cons} = \mathcal{L}\Big(f(\mathcal{F}' \times \mathbf{Q} \mid \theta_t), f(\mathcal{F}'_\gamma \mid \theta_s) \Big) \tag{12}$$

where θ_t, θ_s represent the parameters of heads of teacher model and student model respectively, $\mathcal{L}$ symbolises focal loss [51] or diou loss [52] to guide model to make consistent predictions.

Subsequently, for feature level consistency, we apply several types of simple but effective augmentation such as Gaussian noise and time masking on features. By supervising model to produce the same predictions, the ability of model to extract robust noise-unaware feature representation is improved:

$$\mathcal{L}^f_{cons} = \mathcal{L}\Big(f(\mathcal{F}' \mid \theta_t), f(\tilde{\mathcal{F}}' \mid \theta_s) \Big) \tag{13}$$

Where $\tilde{(\cdot)}$ means applying augmentation to $(\cdot)$. By incorporating these strategies, overall performance of network is improved with enhanced capability in handling action instances of divergent scales.

3.5. Loss Function

For supervised losses, We respectively use focal loss [51] and diou loss [52] for classification error and regression deviation:

$$\mathcal{L}_{sup} = \mathcal{L}^{real}_{focal} + \mathcal{L}^{real}_{diou} \tag{14}$$

where *real* indicates that the training is supervised by real labels.

Unsupervised loss consists of label level consistency loss and feature level consistency loss, given by Equation 12 and 13. Overall loss is calculated by:

$$\mathcal{L} = \mathcal{L}_{sup} + \mathcal{L}^l_{cons} + \mathcal{L}^f_{cons} \tag{15}$$

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Label	Method	Feature	THUMOS14(%)						ActivityNet v1.3(%)			
			0.3	0.4	0.5	0.6	0.7	Avg.	0.5	0.75	0.95	Avg.
10%	ActF + labeled data	I3D	27.7	24.1	19.1	13.8	6.4	18.2	48.9	31.8	3.9	31.0
	ActF + labeled data + pseudo label	I3D	29.4	25.3	19.9	13.5	6.5	18.9	51.0	32.5	4.0	32.1
	ActF + NPL	I3D	32.8	29.6	20.1	11.7	7.2	20.3	51.9	33.4	3.6	32.5
	ActF + NPL*	I3D	34.2	29.0	22.5	15.2	7.2	21.6	51.7	33.2	3.7	32.4
	ActF + MPL(Ours)	I3D	**35.6**	**31.2**	**25.0**	**16.4**	**8.2**	**23.3**	**52.2**	**33.6**	**4.1**	**32.9**
	SSP	TSN	44.2	34.1	24.6	16.9	9.3	25.8	38.9	28.7	8.4	27.6
	SPOT	TSN	49.4	40.4	31.5	22.9	12.4	31.3	49.9	31.1	8.3	32.1
	BMN + SSTAP	TSN	45.6	35.2	26.3	17.5	10.7	27.0	40.7	29.6	**9.0**	28.2
	BMN + NPL	TSN	50.0	41.7	33.5	23.6	13.4	32.4	50.9	32.0	7.9	32.6
	BMN + MPL(Ours)	TSN	**50.5**	**42.2**	**34.0**	**24.2**	**13.4**	**32.9**	**53.4**	**32.6**	7.3	**33.1**
60%	ActF + labeled data	I3D	79.2	73.0	64.4	50.3	36.6	60.7	53.9	36.0	5.0	35.0
	ActF + labeled data + pseudo label	I3D	80.4	74.7	65.8	51.5	37.1	61.9	53.5	35.8	7.5	35.1
	ActF + NPL	I3D	74.5	69.9	62.8	51.1	36.6	59.0	**54.3**	36.7	6.5	35.8
	ActF + NPL*	I3D	80.6	75.6	66.7	53.4	39.3	63.1	54.0	36.6	6.2	35.3
	ActF + MPL(Ours)	I3D	**81.5**	**76.9**	**67.8**	**54.8**	**40.0**	**64.2**	54.1	**36.7**	**8.2**	**35.8**
	SSP	TSN	53.2	46.8	39.3	29.7	19.8	37.8	49.8	34.5	7.0	33.5
	SPOT	TSN	58.9	50.1	42.3	33.5	22.9	41.5	52.8	35.0	8.1	35.2
	BMN + SSTAP	TSN	56.4	49.5	41.0	30.9	21.6	39.9	50.1	34.9	7.4	34.0
	BMN + NPL	TSN	59.0	51.4	**42.9**	34.3	23.3	42.2	53.9	35.8	8.5	35.7
	BMN + MPL(Ours)	TSN	**60.0**	**51.9**	42.5	**34.3**	**23.3**	**42.4**	**54.7**	**36.4**	**8.5**	**35.9**

Table 1: Performance comparison with state-of-the-art methods on THUMOS14 and ActivityNet v1.3. ActF refers to ActionFormer. * means we re-implement the method and conduct experiments under our preconditions. *ActF + labeled data* means in the semi-supervised phase, continue training the model with only labeled data, keeping the same number of epochs with other settings. Here *pseudo label* refers to the paradigm that teacher model produces sharpened predictions to supervise student model where the parameters of teacher are updated by EMA, with no other strategies included.

4. Experiments

4.1. Datasets, Metric and Preconditions

This subsection is to introduce the conditions for experiments.

Datasets. We evaluate our proposed method on the benchmark TAL datasets THUMOS14 [53] and ActivityNet v1.3 [54]. For THUMOS14 [53], following the common setting [26, 27, 6, 55], we use 200 videos in validation set for training and 213 videos in testing set for evaluation. For ActivityNet v1.3 [54], it contains 10k videos for training and 5k videos for evaluation, according to standard practice [6, 55].

Evaluation Metric. We use the mean Average Precision(mAP) as the evaluation metric and report results under specific IoU thresholds. On THUMOS14, the tIoU is set to [0.3:0.7:0.1]. On ActivityNet v1.3, the tIoU for mAP is set to [0.5:0.95:0.05]. Average mAPs are reported based on tIoU, also we report mAP at a tIou of [0.5, 0.75, 0.95].

Dataset Division. Dataset division is essential in SSL. We randomly sampled 10%, and 60% of the videos from the training set to serve as labeled

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PCR	$\mathcal{L}_{cons}^{l}$	$\mathcal{L}_{cons}^{f}$	PAC	10%	60%
✗	✗	✗	✗	18.9(+0.0)	61.9(+0.0)
✓	✗	✗	✗	20.5(+1.6)	62.6(+0.7)
✗	✓	✗	✗	20.6(+1.7)	62.6(+0.7)
✗	✓	✓	✗	21.4(+2.5)	63.4(+1.5)
✓	✓	✓	✗	21.5(+2.6)	63.7(+1.8)
✓	✓	✓	✓	**23.3(+4.4)**	**64.2(+2.3)**

Table 2: Ablation study on effectiveness of each component of the proposed method on THUMOS14, using 10% and 60% labeled videos. The baseline is detector using vanilla pseudo label.

samples, while removing annotations from the remaining videos to treat them as unlabeled samples.

For Fair Comparison. For dataset division, We use the same dataset segmentation as other methods such as SSTAP [4] and SPOT [5] on ActivityNet v1.3. However, there is no publicly available dataset division for THUMOS14, so we conduct the sampling process by ourselves, with the detailed data files accessible. In terms of feature encoding, we use the same pre-extracted TSN [56] feature as SSTAP [4] and SPOT [5] and the same I3D feature as ActionFormer [26]. It should be noticed that previous researches do not release their configures about THUMOS14, so we only compare with NPL [6] when using I3D features on THUMOS14. To align with NPL [6], TSN features are accompanied by BMN [11] network and I3D features are combined with ActionFormer.

4.2. Implementation Details

We use ActionFormer [26] and BMN [11] as the basic frameworks for our method, following NPL [6].

When employing ActionFormer as the framework for video action detection, in the pre-training stage, we only use labeled data to pretrain for 40/15 epochs(for THUMOS14 and ActivityNet v1.3 respectively), the learning rate is set to 1e-4/1e-3, and the learning rate strategy includes 10/5 epochs of warmup and 30/10 epochs of cosine annealing, weight decay is set to 0.05, EMA α is set to 0.995 for the 10% label rate and 0.999 for the 60% label rate. In the semi-supervised stage, we utilize both labeled and unlabeled data for

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Confidence	10%		60%	
	Avg. IoU	Avg. mAP(%)	Avg. IoU	Avg. mAP(%)
All	0.192	18.9	0.289	61.9
Bottom 50%	0.146	18.5	0.192	61.7
Top 50%	**0.243**	**20.5**	**0.386**	**62.6**

Table 3: Ablation study on effectiveness of PCR. The model uses ActionFormer pre-trained with 40 epochs on labeled THUMOS14 data. Average IoU is calculated instance-wise, while average mAP is calculated by using pseudo labels of corresponding parts for semi-supervised training.

training, and set EMA α to 0.999. The remaining training settings remain identical to those used in the pre-training stage.

When using BMN as the base detector, in the pre-training stage, we use labeled data to pretrain for 30/30 epochs with a learning rate of 1e-3/4e-3 and a weight decay of 5e-3/5e-3. Further, we improve the model with pseudo labels for 15 epochs with EMA α been set to 0.99. The temporal dimension of BMN is set to 100/256 for ActivityNet/THUMOS14 respectively.

For key hyper-parameters, N_{psc} is set to 4 with $\alpha_{psc} = 2$. For data augmentation, we apply a 20% time mask, Gaussian noise with standard deviation of 0.01 and temporal left/right shift at 1% likelihood to the features. All pseudo labels are filtered based on a class-confidence threshold of 0.2. When model generates proposals, only regression uncertainty in PCR is used. Besides, SoftNMS is employed for post-processing and experiments are conducted on RTX 2080 Ti $\times$ 1 with fixed random seed.

4.3. Comparison with the State-of-the-Arts

We compare the proposed method MPL with existing main SS-TAL methods in Table 1. Considering the differences in THUMOS14 dataset division and data preprocessing, we reproduce the main methods in NPL [6] for reference and we also present its results in the original paper. It can be seen that there are slight differences in the results. As can be seen, MPL outperforms previous SOTA by a margin of 1.7% and 1.1% in mAP when the label rate is 10% and 60% on THUMOS14, respectively. When it comes to ActivityNet v1.3, MPL demonstrates a advantage of 0.5% over prior methods, showcasing its wide applicability.

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Metric(Top 50%)	10%	60%
	Avg. mAP(%)	Avg. mAP(%)
W/ vanilla pseudo label	18.9	61.9
Action-wise class uncertainty	19.8	62.2
Sample-wise class uncertainty	19.7	62.2
Regression uncertainty	20.3	62.4
All	**20.5**	**62.6**

Table 4: Effectiveness of each component in PCR.

Strategy	10%	60%
	Avg. mAP(%)	Avg. mAP(%)
downsample 2× (interpolation)	19.1	62.0
downsample 2× (index)	19.7	62.2
Gaussian sample $\gamma\times$	**20.6**	**62.6**

Table 5: Effectiveness of Gaussian sampling scaling strategy. The scaling operation directly affects the pre-extracted features.

4.4. Ablation Study

To test the effectiveness of each module in our proposed method, we design the following ablation experiments where ActionFormer [26] works as action detector with I3D [57] features.

Effectiveness of Individual Modules. We started with training using only pseudo labels and gradually added various modules to observe the effect of each module. Experiments are conducted at two tag rates of 10% and 60%, as results can be seen in Table 4. After adding label-level scale consistency loss, the average mAP increased by 1.7% and 0.7%. Then, after applying the feature-level consistency loss for noise-unaware ability, the average mAP further increased by 2.5% and 1.5%. With low-confidence pseudo labels getting corrected, the average mAP finally improved by 4.4% and 2.3%.

Effectiveness of Pseudo-label Confidence Ranking. The PCR module is to estimate confidence of all pseudo labels, so we examine it from two perspectives. One is the effectiveness of its confidence evaluation, which is reflected by the quality and training effect of the high and low confidence pseudo labels it assigns.

We use the average IoU between real intervals and their corresponding

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Confidence	Before PAC	After PAC	only Low conf
All	0.192	0.181↓	**0.207**↑
Top 50%	0.243	0.186↓	0.243
Bottom 50%	**0.146**	**0.178**↑	**0.178**↑

Table 6: The impact of PAC module. PAC has a good improvement effect on low-confidence pseudo labels, but it has a negative effect on high-confidence pseudo labels.

N_{psc}	1	2	4	8	12
Avg. mAP(%)	22.0	23.0	**23.3**	23.3	23.2

Table 7: Hyper-Parameter Selection of N_{psc} at 10% label rate.

pseudo labels to reflect the evaluation effectiveness of pseudo label quality. In TAL, IoU is an efficient indicator for localization precision. As shown in Table 3, low-confidence pseudo labels exhibit poorer localization ability compared to higher ones. Higher confident pseudo labels are accompanied by a 0.097 and 0.194 increase in average IoU at the label rate of 10% and 60%, respectively. In addition, we also conduct experiments to study the contribution of each component, with results shown in Table 4.

Superiority of Gaussian Sampling Scaling Strategy. The DCL part helps the model extract robust temporal-invariant feature representations, thereby considerably enhancing performance on TAL. In this context, we evaluate the performance gain achieved by our proposed temporal scaling strategy in comparison to alternative scaling approaches. Referring to Table 4, in contrast to 2× downsampling via interpolation, our strategy yields a significant improvement of 1.5% and 0.6% in terms of mAP. Additionally, when compared to direct downsampling of the index, we observe a notable improvement of 0.9% and 0.4% in mAP.

Quality Improvements by Pseudo-label Adaptive Correction. The PAC is designed to improve the quality of pseudo labels by refining the predicted temporal boundaries. Our experimental results show that PAC can significantly improve the localization accuracy of low-confidence pseudo labels. At a label rate of 10%, the average IoU improvement for predictions with low confidence can reach more than 22%, as shown in Table 6. If only the unreliable pseudo-labels are corrected, the average IoU of the overall labels will be significantly improved.

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α_{psc}	1	2	3	4	6
Avg. mAP(%)	23.1	**23.3**	23.2	22.9	22.2

Table 8: Hyper-Parameter Selection of α_{psc} at 10% label rate.

Impact of Random Seeds. We select 10 random seeds, ranging from 1234567890 to 1234567899, for a fixed set of settings to observe the impact of random seeds. We find that random seeds can cause the final mAP to fluctuate by up to 0.5%. However, our overall improvement is much higher than this number, thus indirectly reflecting the effectiveness of the MPL method.

Parameter Sensitivity Analysis. we test the impact of N_{psc} and α_{psc} at 10% label rate in Table 7 and 8.

4.5. Visualization

Visualization of PCR. We visualize the classification effect of PCR on pseudo labels. We use a model pre-trained at 60% label rate to generate pseudo labels, and then divide them into two groups with low and high confidence individually. After that, we plot the distribution of these two groups of pseudo labels. As shown in Figure 4, the classification effect is so obvious that the high-confidence group has an average IoU of 0.386 and an average category accuracy of 87.0% while the low-confidence group has an average IoU of 0.192 and an average category accuracy of 53.8%.

Visualization of PAC. We also visualize the repair effect of PAC on low-confidence pseudo labels. Here, we repaired the pseudo labels under the precondition that the model is trained with 10% labeled video data. As shown in Figure 5, the distribution center of low-confidence pseudo labels shifted towards the upper right corner, which means that the pseudo labels became more accurate. In fact, the average IoU increased from 0.146 to 0.178, while the average accuracy increased from 41.72% to 46.47%.

Visualization of Scale-invariant Learning. Our Dual-level Consistency Learning (DCL) can alleviate the scale deviation in predictions caused by inappropriate downsampling strategies. Meanwhile, compared to not using temporal scaling, our DCL can make the model's prediction of action length more stable and accurate. To demonstrate this superiority, we set a simple indicator LR, which is the ratio of the length of the prediction interval to the length of the GT interval. More specifically, we calculate the

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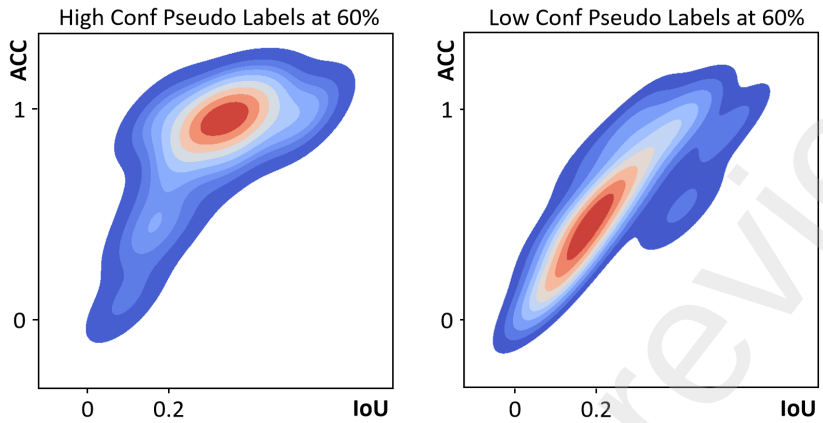

Figure 4: The distribution of pseudo labels with high confidence (left) and low confidence (right) at 60% label rate.

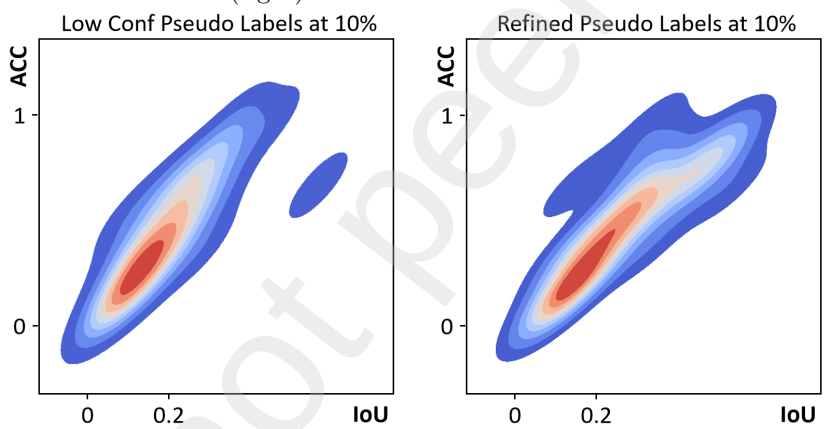

Figure 5: The distribution of low-confidence pseudo labels (left) and refined pseudo labels (right) at 10% label rate.

LRs at instance level for all actions, log-normalize the LRs and estimate corresponding kernel density function curve, as shown in Figure 6.

5. Conclusion

In this paper, we propose a novel framework called MPL that consists of three modules to mine the pseudo labels with low confidence for SS-TAL. Concretely, we first design an overall evaluating indicator for pseudo labels of certain samples. The indicator takes into account the category uncertainty at the action level and at the sample level, as well as the instability of

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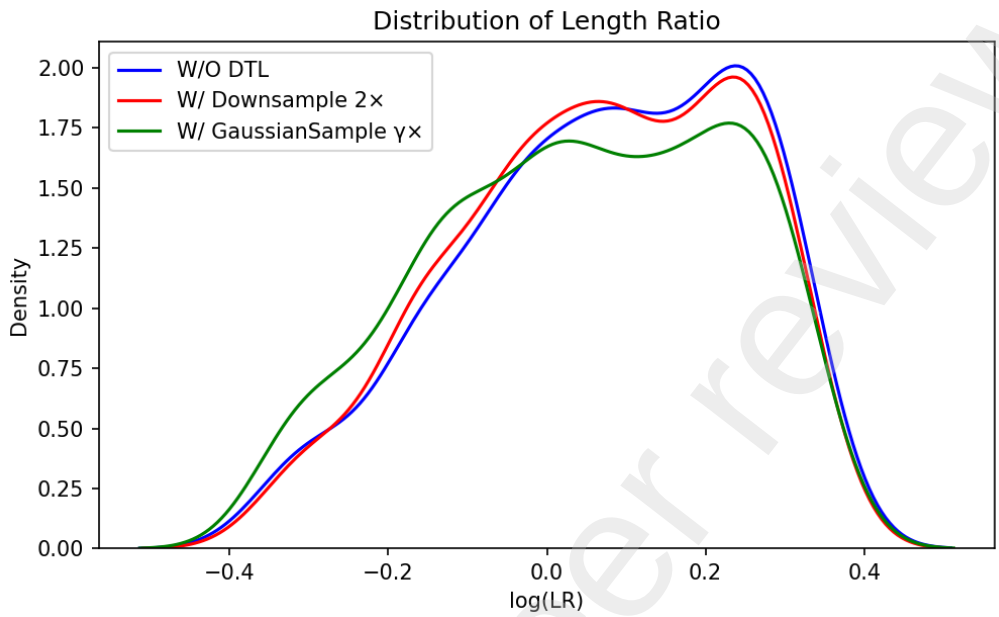

Figure 6: The distribution of log-normalized length ratio. The positive skewed distribution reflected by the red and blue curves indicates that the predicted intervals are generally larger. Green curve demonstrates that DCL with Gaussian sampling scaling strategy has significantly mitigated such negative tendency.

predicted boundaries. Then, pseudo label adaptive correction is applied to low-confidence labels obtained by the teacher model, to improve the precision of temporal action localization. Finally, dual-level consistency learning helps the model to learn scale-invariant and noise-unaware feature representation, using the Gaussian sampling scaling strategy. Extensive experiments show the effectiveness of the proposed MPL on two benchmark datasets: THUMOS14 and ActivityNet v1.3.

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